

Datafort Apache mTLS Setup

Installation and verification instructions for the client-certificate trust anchor on **datafort.folksfirstlabs.com**.

Environment	cPanel/WHM - Apache 2.4
cPanel user	folksf1
Required attachment	company-ca.crt
Rollout state	SSLVerifyClient optional

SECURITY BOUNDARY

The server receives only the public certificate **company-ca.crt**. Do not request, upload, or install **company-ca.key** or its passphrase. The private key remains offline with the certificate owner.

Prepared 26 August 2026

1. Install the supplied CA certificate

Upload the attached **company-ca.crt** to the server, then run the following as root. Replace the temporary upload path with the actual uploaded location.

```
mkdir -p /etc/ssl/datafort

cp /path/to/uploaded/company-ca.crt \
/etc/ssl/datafort/company-ca.crt

chown root:root /etc/ssl/datafort/company-ca.crt
chmod 644 /etc/ssl/datafort/company-ca.crt
```

Verify its identity before changing Apache:

```
openssl x509 -in /etc/ssl/datafort/company-ca.crt \
-noout -subject -issuer -dates -fingerprint -sha256
```

EXPECTED CERTIFICATE

Subject and issuer: **CN=Movenetics Digital Device CA**

Valid through: **23 August 2036**

SHA-256: 26:E8:2C:E8:25:94:2D:18:18:53:CA:B8:1E:B8:FF:AF:

B5:0E:FD:D1:97:EC:6B:D7:16:EF:C3:44:80:B1:79:09

2. Create the cPanel per-vhost include

Use cPanel's userdata include path so the configuration survives AutoSSL and cPanel virtual-host rebuilds.

```
mkdir -p /etc/apache2/conf.d/userdata/ssl/2_4/folksfi1/\
datafort.folksfirstlabs.com

nano /etc/apache2/conf.d/userdata/ssl/2_4/folksfi1/\
datafort.folksfirstlabs.com/datafort-mtls.conf
```

3. Add the client-certificate directives

Place exactly the following in **datafort-mtls.conf**. Do not wrap these lines in another <VirtualHost> block; cPanel includes this file inside the existing SSL virtual host.

```
# Datafort client-certificate authentication

SSLCACertificateFile /etc/ssl/datafort/company-ca.crt
SSLVerifyDepth 2

# Keep optional during laptop enrolment.
SSLVerifyClient optional

# Expose the certificate result to PHP.
SSLOptions +StdEnvVars +ExportCertData
```

DO NOT MODIFY AUTOSSL

Do not change SSLCertificateFile, SSLCertificateKeyFile, or SSLCertificateChainFile. Those directives belong to the website's server certificate and remain managed by cPanel AutoSSL.

4. Rebuild and validate Apache

```
/scripts/ensure_vhost_includes --user=folksfi1
apachectl configtest
```

Continue only when the output is:

```
Syntax OK
```

```
/scripts/restartsrv_httpd
```

The CA file is read when Apache starts, so a restart is required after installation.

Confirm the include and virtual host:

```
apachectl -t -D DUMP_INCLUDES | grep -i datafort
apachectl -t -D DUMP_VHOSTS
```

5. Verify PHP receives the TLS result

Temporarily create **mtls-test.php** in the Datafort document root:

```
/home/folksfi1/datafort.folksfirstlabs.com/mtls-test.php

<?php
header('Content-Type: text/plain');

foreach ($_SERVER as $key => $value) {
    if (strpos($key, 'SSL_') === 0) {
        echo $key . ' = ' . $value . "\n";
    }
}

echo "\n--- verdict ---\n";
echo isset($_SERVER['SSL_CLIENT_VERIFY'])
? "OK: Apache is exposing client-certificate variables.\n"
: "NOT WORKING: SSL variables are unavailable.\n";
```

Open:

```
https://datafort.folksfirstlabs.com/mtls-test.php
```

Expected before any laptop certificate is installed:

```
SSL_CLIENT_VERIFY = NONE

--- verdict ---
OK: Apache is exposing client-certificate variables.
```

NONE IS CORRECT AT THIS STAGE

It means the browser supplied no client certificate, while Apache still exposed the verification result to PHP. If the variable is absent, confirm the include loaded and that `SSLOptions +StdEnvVars +ExportCertData` is active.

Delete the public diagnostic immediately after the check:

```
rm /home/folksfi1/datafort.folksfirstlabs.com/mtls-test.php
```

6. Keep the safe rollout state

DO NOT ENABLE REQUIRE YET

Leave `SSLVerifyClient optional` in place until authorized laptop certificates have been issued, installed, and successfully tested. Changing it to `require` now will reject every laptop that does not already hold a valid certificate.

Stage	Apache setting	Result
Current	<code>SSLVerifyClient optional</code>	Requests a certificate but permits missing certificates.
Later	<code>SSLVerifyClient require</code>	Rejects connections without a trusted client certificate.

7. Return this deployment evidence

- 1 **Output of `apachectl configtest`.**
- 2 **Confirmation that the cPanel userdata include loaded.**
- 3 **Output from `mtls-test.php`.**
- 4 **Confirmation that `mtls-test.php` was deleted.**
- 5 **Confirmation that `SSLVerifyClient` remains `optional`.**
- 6 **Confirmation that only `company-ca.crt` was installed - never the CA private key.**

Authoritative reference

Apache HTTP Server 2.4 `mod_ssl` documentation: https://httpd.apache.org/docs/2.4/mod/mod_ssl.html